

**Sec: NAITS**  
**Time: 3 Hrs.**

**PTA – 01**

**Date: 04-02-2024**  
**Max. Marks: 300**

## **JEE-ADVANCE-2018-P2-Model**

**Time: 3:00 Hour's**

### **IMPORTANT INSTRUCTIONS**

**Max Marks: 180**

### **PHYSICS:**

| Section                | Question Type                                                                      | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|------------------------|------------------------------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 1 – 6)   | Questions with Multiple Correct Choice (partial marking scheme) (+1,0)             | +4        | -2         | 6         | 24          |
| Sec – II(Q.N : 7 – 14) | Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) | +3        | 0          | 8         | 24          |
| Sec – III(Q.N : 15-18) | Matrix Matching Type                                                               | +3        | -1         | 4         | 12          |
| <b>Total</b>           |                                                                                    |           |            | <b>18</b> | <b>60</b>   |

### **CHEMISTRY:**

| Section                | Question Type                                                                      | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|------------------------|------------------------------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 19 – 24) | Questions with Multiple Correct Choice (partial marking scheme) (+1,0)             | +4        | -2         | 6         | 24          |
| Sec – II(Q.N : 25 –32) | Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) | +3        | 0          | 8         | 24          |
| Sec – III(Q.N : 33-36) | Matrix Matching Type                                                               | +3        | -1         | 4         | 12          |
| <b>Total</b>           |                                                                                    |           |            | <b>18</b> | <b>60</b>   |

### **MATHEMATICS:**

| Section               | Question Type                                                                      | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|-----------------------|------------------------------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N:37 – 42)  | Questions with Multiple Correct Choice (partial marking scheme) (+1,0)             | +4        | -2         | 6         | 24          |
| Sec – II(Q.N :43–50)  | Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) | +3        | 0          | 8         | 24          |
| Sec –III(Q.N : 51-54) | Matrix Matching Type                                                               | +3        | -1         | 4         | 12          |
| <b>Total</b>          |                                                                                    |           |            | <b>18</b> | <b>60</b>   |

| Exam Syllabus      |                                                                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PHYSICS</b>     | Units and Measurement, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy and Power, System of Particles and Rotational Motion          |
| <b>CHEMISTRY</b>   | Redox reaction, Some Basic concept in Chemistry, Atomic Structure, Periodic Properties Chemical bonding and molecular structure States of Matter Thermodynamics |
| <b>MATHEMATICS</b> | Sequence and Series Quadratic Equations Binomial theorem Permutation and Combination Trigonometric functions Trigonometric Equations                            |

| Subject       | Physics         | Chemistry   | Mathematics        |
|---------------|-----------------|-------------|--------------------|
| Setter Branch | AURANGABAD      |             |                    |
| SME Name      | Mr. ABDUL WAHAB | Mr. D. RAO  | MR. PRADEEP MISARA |
| Phone Number  | 93404 77855     | 99874 27722 | 93736 90662        |

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*Rough Work*

## PHYSICS

Max. Marks: 60

SECTION – I  
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE than ONE** option can be correct.

**Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.**

1. The mass, length and velocity in two systems of units are related as under:

$$\text{i) } m' = \left(\frac{\alpha}{\beta}\right)m \quad \text{ii) } l' = (\alpha\beta)l \quad \text{iii) } v' = \left(\frac{\alpha^2}{\beta}\right)v$$

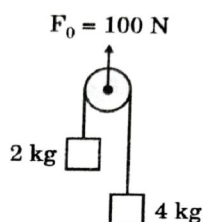
Where  $m', l'$  and  $v'$  belong to one system and  $m, l, v$  belong to the other system. Here  $\alpha$  and  $\beta$  are dimensionless constants. Which of the following is/are correct?

$$\text{A) } a' = \left(\frac{\alpha^3}{\beta^3}\right)a \quad \text{B) } F' = \left(\frac{\alpha^2}{\beta^2}\right)F \quad \text{C) } T' = \left(\frac{\beta^2}{\alpha^2}\right)T \quad \text{D) } \rho' = \left(\frac{\alpha^2}{\beta^2}\right)\rho, \rho \text{ is momentum.}$$

2. A particle is moving along X-axis whose position is given by  $x = 4 - 9t + \frac{t^3}{3}$ . Mark the correct statement (s) in relation to its motion.

- A) Direction of motion is not changing at any of the instants  
 B) Direction of motion is changing at  $t = 3\text{s}$   
 C) For  $0 < t < 3\text{s}$ , the particle is slowing down  
 D) For  $0 < t < 3\text{s}$ , the particle is speeding up

3. A block masses  $m_1 = 2\text{ kg}$  and  $m_2 = 4\text{ kg}$  hang over a massless pulley as shown in the figure. A force  $F_0 = 100\text{ N}$  acting at the axis of the pulley accelerates the system upwards. Then

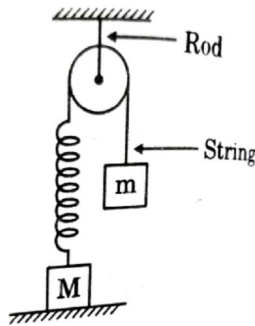


- A) Acceleration of 2 kg mass is  $15\text{ m/s}^2$       B) Acceleration of 4 kg mass is  $2.5\text{ m/s}^2$   
 C) Acceleration of both the masses is same      D) Acceleration of both the masses is upward

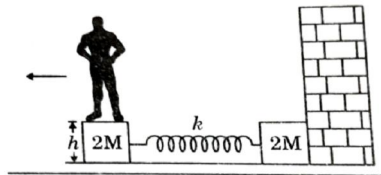
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4. In figure, a block of mass  $m$  is released from rest when spring was in its natural length. The pulley also has mass  $m$  but it is frictionless. Suppose the value of  $m$  is such that finally it is just able to lift the block  $M$  up after releasing it.



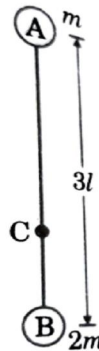
- A) The weight of  $m$  required to just lift  $M$  is  $\frac{M}{2}g$
- B) The tension in the rod, when  $m$  has zero acceleration  $\frac{M}{2}g$
- C) The normal force acting on  $M$  when  $m$  has zero acceleration  $\frac{M}{2}g$
- D) The tension in the string when displacement of  $m$  is maximum possible is  $Mg$
5. Two identical blocks of mass  $2M$  are joined by means of a light spring of spring constant  $k$ . A man of mass  $M$  is standing on one of the block as shown in the diagram. If man jumps horizontally with a velocity  $V$  relative to block and horizontal surface is smooth, then



- A) The maximum compression in the spring is  $\sqrt{\frac{2M}{k}} \left( \frac{V}{3} \right)$
- B) Man lands at horizontal distance  $V \sqrt{\frac{2h}{g}}$  from initial position of the block
- C) Right block loses contact with wall when the elongation in spring is maximum
- D) Velocity of centre of mass of two blocks after  $2M$  loses contact with wall is  $\frac{V}{6}$

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6. A light rigid rod AB of length  $3l$  has a point mass  $m$  at end A and a point mass  $2m$  at end B. It is kept on a smooth mass of the system. Initially the system is at rest. The mass  $1m$  is suddenly given a velocity  $v_0$  towards right. Take Z axis to be perpendicular to the plane of the paper. [Points C is center of mass ]



- A) The minimum moment of inertia ( about Z-axis),  $I_{zz}$  of the system is  $5ml^2$ .
- B) The magnitude of tension in the rod in subsequent motion is  $\frac{2mv_0^2}{9l}$
- C) The ratio of moment of inertia about Z-axis at points A and B ,  $\frac{I_{zz}^A}{I_{zz}^B} = 2$
- D) Point C remains stationary during subsequent motion.

### SECTION-II (NUMERICAL VALUE TYPE)

This section contains 8 questions. Each question is numerical value. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place.

(e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

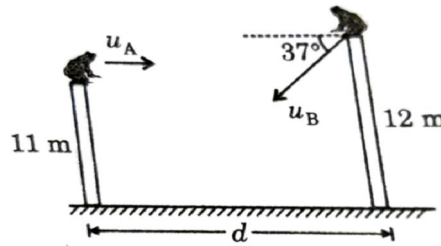
**Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases.**

7. A train is moving along a straight line with a constant acceleration 'a'. A boy standing in the train throws a ball forward with a speed of 10 m/s , at an angle of  $60^\circ$  to the horizontal. The boy has to move forward by 1.16 m inside the train to catch the ball back at the initial height. The acceleration of the trains, in  $m/s^2$ , is

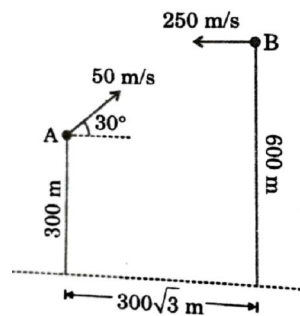
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8. Two frogs A and B jump simultaneously from top of two poles 11 m and 12 m high respectively towards each other during a fight as shown in figure. They collide in mid-air. If the velocities  $u_A$  and  $u_B$  are 2m/s and 5m/s respectively, the value of separation  $d$ , in meters is

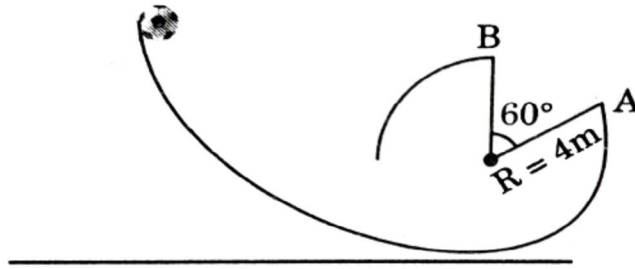


9. A box of mass 1 kg is lying on a rough horizontal floor and a horizontal force acting on box has a magnitude of  $F = 3t^2 N$ , where  $t$  is in seconds. If the box starts from rest, determine its speed ( in m/s) when  $t = 3$  sec. The coefficient of static and kinetic friction between the box and horizontal floor are  $\mu_s = 0.3$  and  $\mu_k = 0.2$ .
10. Two particles A and B are projected from top of two towers of height 300 m and 600 m with velocity 50 m/s and 250 /s respectively . The angular velocity of B. w.r.t A is  $\alpha$  rad/ min. find the value of  $10\alpha$

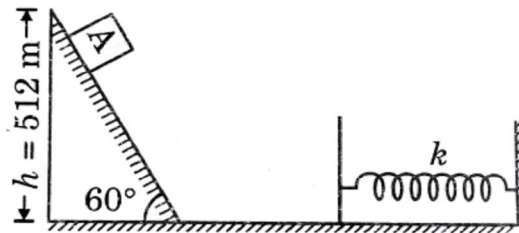


*Rough Work*

11. As shown in the figure, a ball is released from a certain height which has to perform circular motion on the vertical track of radius 4m. The track is absent between point A and B. Calculate the height ( in m) from where the ball to be released so that it will reach at highest point B of the circular track.

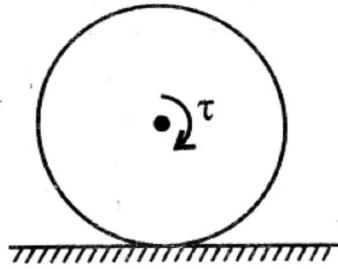


12. A block A is released from the top of a fixed wedge at height  $h = 512$  m. All surfaces are frictionless. A collides with spring elastically while moving from inclined to plane surface or vice-versa and A never loses contact with surface. Collision between block and horizontal surface/incline plane are perfectly inelastic. Find the height ( in meter) attained by block when it climbs the wedge second time

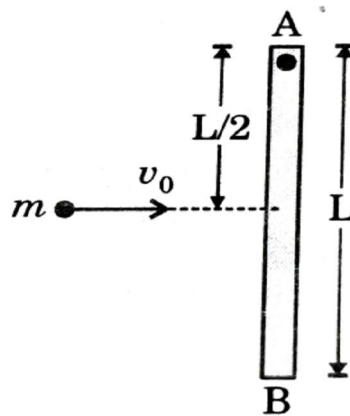


*Rough Work*

13. A wheel of mass  $m$  can be assumed to be a ring. Its radius is  $R$ . It is on a level ground. An external torque is applied to it about its axis.  $\tau = \frac{mgR}{2}$ . If it rolls purely, what is the friction force exerted by the ground. (Take  $g = 10 \text{ m/s}^2$ )



14. A uniform rod AB which is free to swing in the vertical plane about a horizontal axis through A, is hanging freely. A particle of equal mass strikes the rod with a velocity  $v_0$  and gets stuck to it. Find the angular velocity (in rad/sec) of the combination immediately after the collision. (Given  $v_0 = 14 \text{ cm/s}$ ,  $L = 12 \text{ cm}$ )




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*Rough Work*

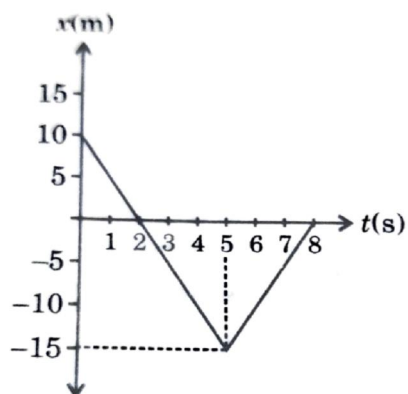


### SECTION-III (MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

15. Consider a particle moving in a straight line



| COLUMN – I |                                         | COLUMN – II |          |
|------------|-----------------------------------------|-------------|----------|
| A)         | Distance travelled in 1 to 7 sec        | P)          | 10 unit  |
| B)         | Magnitude of displacement in 1 to 7 sec | Q)          | 30 units |
| C)         | Average acceleration between 4 to 6 sec | R)          | 0        |
| D)         | Change in speed between 1 to 6 sec      | S)          | 5 unit   |

A) A - R ; B - P ; C - Q ; D - S

B) A - R ; B - P ; C - S ; D - Q

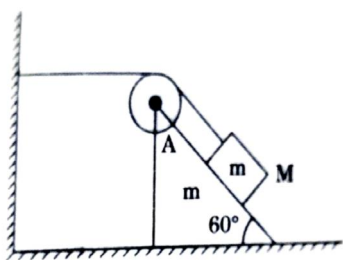
C) A - Q ; B - P ; C - S ; D - R

D) A - R ; B - S ; C - P ; D - Q

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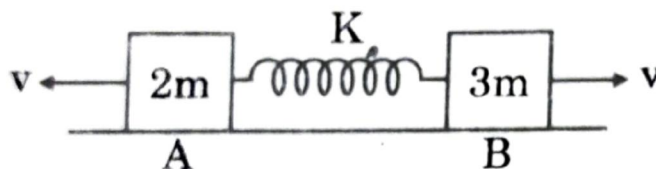
*Rough Work*

16. In the following arrangement string is light and inextensible. All surfaces are smooth. Treating  
 $a$ =acceleration of wedge  $b$ =acceleration of block B  
 $T$ = tension in the string  $N$ = Normal reaction on block B applied by wedge A



| COLUMN – I |                 | COLUMN – II |             |
|------------|-----------------|-------------|-------------|
| A)         | $\frac{4a}{g}$  | P)          | $\sqrt{3}$  |
| B)         | $\frac{8T}{mg}$ | Q)          | $3\sqrt{3}$ |
| C)         | $\frac{8N}{mg}$ | R)          | 1           |
| D)         | $\frac{3b}{a}$  | S)          | 3           |

- A) A -R ; B -P ; C -Q ; D -S B) A -P ; B -Q ; C -R ; D -S  
 C) A - Q; B - Pp; C -s ; D - r D) A -R ; B -S ; C -P; D -Q
17. Two blocks A and B of masses  $2m$  and  $3m$  placed on smooth horizontal surface are connected with a light spring. The two blocks are gives velocities as shown when spring is at natural length.



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| COLUMN – I |                                                                                           | COLUMN – II |                |
|------------|-------------------------------------------------------------------------------------------|-------------|----------------|
| A)         | Minimum magnitude of velocity of A ( $v_{A_{\min}}$ ) during motion                       | P)          | $v$            |
| B)         | Maximum magnitude of velocity of A ( $v_{A_{\max}}$ ) during motion                       | Q)          | $\frac{v}{5}$  |
| C)         | Maximum magnitude of velocity B ( $v_{B_{\max}}$ ) during motion                          | R)          | 0              |
| D)         | Velocity of centre of mass ( $v_{CM}$ ) of the system comprised of blocks A, B and spring | S)          | $\frac{7v}{5}$ |

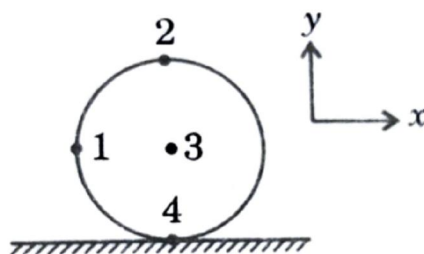
A) A -R ; B -S ; C -P ; D -Q

B) A -R ; B -P ; C -S ; D -S

C) A - Q; B - R; C -P ; D - S

D) A -R ; B -P ; C -P; D -R

18. A rigid cylinder is kept on a smooth horizontal surface as shown. If column – A indicates velocities of various points (3-centre of cylinder, 2-top point, 4-bottom point, 1-on the level of 3 at the rim) on it shown, choose correct state of motion from Column B.




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| COLUMN – I |                                                       | COLUMN – II |                                   |
|------------|-------------------------------------------------------|-------------|-----------------------------------|
| A)         | $\vec{v}_1 = \hat{i} + \hat{j}, \vec{v}_2 = 2\hat{i}$ | P)          | Pure rotation about centre        |
| B)         | $\vec{v}_1 = \hat{i} + \hat{j}, \vec{v}_3 = -\hat{i}$ | Q)          | Rolling without slipping to left  |
| C)         | $\vec{v}_2 = \hat{i}, \vec{v}_3 = 0$                  | R)          | Rolling without slipping to right |
| D)         | $\vec{v}_4 = 0, \vec{v}_1 = -\hat{i} - \hat{j}$       | S)          | Not possible                      |

A) A – Q; B – R; C – Q ; D –S

B) A – R; B – Q; C – P ; D –S

C) A – P; B – Q; C – R ; D –S

D) A – R; B – S; C – P ; D –Q

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*Rough Work*

## CHEMISTRY

Max.Marks:60

SECTION – I  
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE than ONE** option can be correct.

**Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.**

19. Radiation of wavelength  $200 \text{ \AA}$  falls on a platinum surface. If the work function of the metal is  $5 \text{ eV}$ . Which of the following results are correct about experiment ?
- (A) The velocity of photoelectrons increases with increase in intensity of radiation  
(B) Photo-emission of electrons takes place  
(C) the threshold frequency of the metal is  $1.21 \times 10^{15} \text{ sec}^{-1}$  ( approximately)  
(D) the velocity of the photo-electrons is  $4.48 \times 10^6 \text{ m/sec}$ . ( approximately)
20. A mixture of  $\text{CH}_4$  and  $\text{C}_2\text{H}_2$  occupied a certain volume at a total pressure equal to 63 torr. The same gas mixture was burnt to  $\text{CO}_2$  and  $\text{H}_2\text{O} (\ell)$ . The  $\text{CO}_2 (\text{g})$  alone was collected in the same volume and at the same temperature, the pressure was found to be 69 torr. What was the mole fraction of  $\text{CH}_4$  in the original gas mixture ?
- (A)  $\frac{19}{21}$                       (B)  $\frac{19}{20}$                       (C)  $\frac{17}{18}$                       (D)  $\frac{15}{16}$
21. Select equations having endothermic step -
- (A)  $\text{S}^- (\text{g}) \rightarrow \text{S}^{2-} (\text{g})$                       (B)  $\text{Na}^+ (\text{g}) + \text{Cl}^- (\text{g}) \rightarrow \text{NaCl}(\text{s})$   
(C)  $\text{Na}(\text{g}) \rightarrow \text{Na}^+ (\text{g})$                       (D)  $\text{Al}^{2+} (\text{g}) \rightarrow \text{Al}^{3+} (\text{g})$
22. Enthalpy of atomisation of  $\text{C}_2\text{H}_6 (\text{g})$  and  $\text{C}_3\text{H}_8 (\text{g})$  are  $620 \text{ kJ/mol}$  and  $880 \text{ kJ/mol}$ . Therefore ( $\Delta$  means bond dissociation enthalpy)
- (A)  $\Delta_{\text{C-H}}$  is  $90 \text{ kJ/mol}$   
(B)  $\Delta_{\text{C-C}}$  is  $80 \text{ kJ/mol}$   
(C)  $\Delta_{\text{C=C}}$  is  $160 \text{ kJ/mol}$   
(D) Enthalpy of atomisation of  $\text{CH}_4$  is approximately  $360 \text{ kJ/mol}$ .

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*Rough Work*

23. When 1.3 mole of butane ( $C_4H_{10}$ ) is burned in excess of oxygen. What is true about this -  
( molar volume at STP = 32.4 Lit)  
(A) Volume of  $CO_2$  produced at S.T.P. is 116.48 L  
(B) Volume of  $CO_2$  produced at S.T.P. is 89.6 L  
(C) Mass of  $H_2O$  produced is 117 g  
(D) Mass of water produced is 217 g
24. Correct statement is/are  
(A) In  $Li_2$ ,  $B_2$ ,  $C_2$  molecules, 2s, 2p atomic orbitals mixing is present in the formation of molecular orbitals  
(B) In  $C_2$  molecule, double bond in between carbon atoms consist two  $\pi$  - bonds.  
(C)  $B_2$  and  $O_2$  molecules are para magnetic  
(D) For conversation of  $O_2 \rightarrow O_2^+$  bond length decreases.

**SECTION-II**  
**(NUMERICAL VALUE TYPE)**

This section contains 8 questions. Each question is numerical value. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place.

(e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

**Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases.**

25. Number of hydrogen bonds present in a molecule of  $CCl_3CH(OH)_2$  (chloral hydrate) is
26. If for an element (X), the value of successive ionisation energies  $I_1$ ,  $I_2$ ,  $I_3$ ,  $I_4$  and  $I_5$  are 800, 2427, 3658, 25024, 32824  $KJ\ mol^{-1}$  respectively, then the number of valence electron present are :-  
Determine the radius of  $F^-$  in pm in nearest possible integer.
27.  $NH_3$  is produced according to the following reaction :  
 $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$   
In an experiment 0.25 mol of  $NH_3$  is formed when 0.5 mol of  $N_2$  is reacted with 0.5 mol of  $H_2$ . What is % yield?
28. Eq. wt. of chlorine molecule in the equation  
 $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$

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*Rough Work*

29. At what temperature most probable speed of  $O_2$  molecules have the same value of root mean square speed of  $O_2$  molecules at 300 K ?
30. A gas is heated in a cylinder fitted with a nozzle from  $27^\circ\text{C}$  for 20 minutes. It is found that  $\frac{2}{3}$ rd of the original gas is diffused out through the nozzle. What would be the difference in temperature (give your answer in terms of multiples of 100K) between initial and final states ?
31.  $\Delta H_f^0$  for  $Ca^{2+}(aq)$ ,  $CO_3^{2-}(aq)$  and  $CaCO_3(s)$  are  $-129.80$ ,  $-161.65$  and  $-288.45 \text{ kcal/mol}$  respectively. The enthalpy change when infinitely dilute solutions of  $CaCl_2$  and  $Na_2CO_3$  are mixed is 'x' kcal/mol. Then x is
32. What will be the atomic number of an atom if its electronic configuration is  $(n-2)s^2(n-1)s^a p^b ns^2 p^2$  where  $n = 3$ ,  $a = 2$ , and  $b = 6$  ?

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*Rough Work*

### SECTION-III (MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

33. Match the Following: -

Column I

(A)  $\text{CN}^-$

(B)  $\text{C}_2$

(C) NO

(D)  $\text{Ne}_2^+$

Column II

(P) Bond order is 2, diamagnetic

(Q) Bond order is 3, diamagnetic

(R) Bond order is  $1/2$ , paramagnetic

(S) Bond order is  $5/2$ , paramagnetic

A) A – P; B – P ; C – Q; D – R

B) A- Q; B –P ; C- S; D –R

C) A – S; B – P; C- Q; D – R

D) A – P; B – Q; C – R; D- S

34. Match the Following: -

|     | Column – I |     | Column – II         |
|-----|------------|-----|---------------------|
| (A) | $T_c/P_c$  | (P) | 2                   |
| (B) | $T_c/V_c$  | (Q) | $a/Rb$              |
| (C) | $T_B$      | (R) | $8b/R$              |
| (D) | $T_i/T_B$  | (S) | $\frac{8a}{81Rb^2}$ |

A) A – P; B – P ; C – Q; D – R

B) A- R; B –S ; C- Q; D – P

C) A – S; B – P; C- Q; D – R

D) A – P; B – Q; C – R; D- S

35. Match the Following: -

|     | Column – I                     |     | Column –          |
|-----|--------------------------------|-----|-------------------|
| (A) | Rutherfordium -(At. No. = 104) | (P) | Period number = 7 |
| (B) | Rontgenium - (At. No. = 111)   | (Q) | Group number = 4  |
| (C) | Thorium - (At. No. = 90)       | (R) | d-block elements  |
| (D) | Neptunium                      | (S) | f-block elements  |

A) A – P,Q,R ; B – P,R ; C – P,S ; D – P,S

B) A- R; B –S ; C- Q; D – P

C) A – S; B – P; C- Q; D – R

D) A – P; B – Q; C – R; D- S

*Rough Work*



## 36. Match the Following: -

|     | Column – I                                                                                            |     | Column – II                                                       |
|-----|-------------------------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------|
| (A) | $\text{CO(g)} + \frac{1}{2} \text{O}_2 \text{(g)} \rightarrow \text{CO}_2 \text{(g)}$                 | (P) | Standard enthalpy of formation of $\text{CO}_2$                   |
| (B) | $\text{C(graphite)} + \text{O}_2 \text{(g)} \rightarrow \text{CO}_2 \text{(g)}$                       | (Q) | Enthalpy of reaction                                              |
| (C) | $\text{H}_2 \text{(g)} + \frac{1}{2} \text{O}_2 \text{(g)} \rightarrow \text{H}_2\text{O} \text{(l)}$ | (R) | Enthalpy of combustion of $\text{CO} \text{(g)}$                  |
| (D) | $2\text{H}_2 \text{(g)} + \text{O}_2 \text{(g)} \rightarrow 2\text{H}_2\text{O} \text{(g)}$           | (S) | Standard enthalpy of formation of $\text{H}_2\text{O} \text{(l)}$ |

A) A – P; B – P ; C – Q,P; D – R,S

B) A- S; B – R; C- Q; D – P

C) A – R,Q; B – P,Q; C- S,Q; D – Q

D) A – P; B – Q; C – R; D- S

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*Rough Work*

## MATHEMATICS

Max.Marks:60

**SECTION- I**  
**(Maximum Marks : 24)**

This section contains SIX (06) questions.

Each question has FOUR options for correct answer(s). ONE OR MORE THAN ONE of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

**Full Marks :** +4 If only (all) the correct option(s) is (are) chosen.

**Partial Marks :** +3 If all the four options are correct but ONLY three options are chosen.

**Partial Marks :** +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

**Partial Marks :** +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

**Zero Marks :** 0 If none of the options is chosen (i.e. the question is unanswered).

**Negative Marks:** -2 In all other cases.

37. A man moves one unit distance for each step he takes. He always moves either parallel to the x-axis or y-axis. He starts at the point (0,0) and reaches the point (1,1) at the end of six steps. The number of ways he can do it is  $n$ . Then the number of positive integral solutions of  $x_1x_2x_3 = n - 270$  is  
 A) 25                      B) 27                      C) 26                      D) 28
38. Let  $a, b, c, d, e \in N$  such that  $a + b + c + d + e = p^3$  and  $b + c + d = q^2$  where  $p, q \in N$ . If  $a, b, c, d, e$  (in order) are in arithmetic sequence with common difference unity, then find the least value of  $(p^3 + q^2)$ .  
 A) 2250                      B) 9450                      C) 5400                      D) 60
39. If the number of ordered pairs  $(a, b)$  ( $a, b$  are integers) such that  
 $\min(x^2 + (a - b)x + (1 - a - b)) > \max(-x^2 + (a + b)x - (1 + a + b)) \forall x \in R$  is  $n$  then  $n$  is divisible by  
 A) 3                      B) 6                      C) 9                      D) 12

---

*Rough Work*

40. If  $b > a > 0$ , then the expression E given by

$$E = \frac{1}{\sqrt{b-a}} \frac{\sqrt{\frac{b-a}{a}} \sin x}{\sqrt{1 + \left(\sqrt{\frac{b-a}{a}} \sin x\right)^2}} \cdot \sqrt{a + b \tan^2 x} \text{ must be equal to}$$

- A)  $\tan x$                       B)  $|\tan x|$                       C)  $\frac{\sin x}{|\cos x|}$                       D) unity if  $x = \frac{11\pi}{4}$
41. Let  $f(x) = 1 + x^{111} + x^{222} + x^{333} + \dots + x^{999}$  then  $f(x)$  is divisible by
- A)  $x+1$                       B)  $x$                       C)  $x-1$                       D)  $1 + x^{222} + x^{444} + x^{666} + x^{888}$
42. Let  $f(\theta) = \left(\cos \theta - \cos \frac{\pi}{8}\right) \left(\cos \theta - \cos \frac{3\pi}{8}\right) \left(\cos \theta - \cos \frac{5\pi}{8}\right) \left(\cos \theta - \cos \frac{7\pi}{8}\right)$  then
- A) maximum value of  $f(\theta) \forall \theta \in R$  is  $\frac{1}{4}$
- B) maximum value of  $f(\theta) \forall \theta \in R$  is  $\frac{1}{8}$
- C)  $f(0) = \frac{1}{8}$
- D) Number of principle solutions of  $f(\theta) = 0$  is 8

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*Rough Work*

**SECTION - II**  
**(Maximum Marks : 24)**

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.

Answer to each question will be evaluated according to the following marking scheme:

**Full Marks:** +3 If ONLY the correct numerical value is entered as answer.

**Zero Marks:** 0 In all other cases.

43. Four persons A, B, C, D are to be seated in a row such that B does not follow A, C does not follow B and D does not follow C. Then the number of ways of selecting them is  $l$  then  $\left\lceil \frac{l}{10} \right\rceil =$
44. If  $\cos \theta + \cos^2 \theta + \cos^3 \theta = 1$  and  $\sin^6 \theta = a + b \sin^2 \theta + c \sin^4 \theta$  then  $a + b + c =$
45. The value of  $4 \sin 50^\circ - \sqrt{3} \tan 50^\circ = \dots$
46. Let  $f(n) = \sum_{r=0}^n \sum_{k=r}^n \binom{k}{r} C_r$  find the total number of divisions of  $f(9)$
47. If  $5^{81} + \lambda$  is divisible by 26, then minimum positive value of  $\frac{\lambda}{3}$  must be
48. The sum of the series  $3 \cdot {}^{2007}C_0 - 8 \cdot {}^{2007}C_1 + 13 \cdot {}^{2007}C_2 - 18 \cdot {}^{2007}C_3 + \dots$  upto 2008 terms is K, then k is
49. The number of real values of 'a' for which the largest value of the function  $f(x) = x^2 + ax + 2$  in the interval  $[-2, 4]$  is 6 will be
50. If  $x + \sin y = 2014$  and  $x + 2014 \cos y = 2013$ ,  $0 \leq y \leq \frac{\pi}{2}$ , then find the value of  $[x + y] - 2005$  (where  $[.]$  denotes greatest integer function)

*Rough Work*

**SECTION-III**  
**(MATCHING LIST TYPE)**

This section contains 4 questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

51.

| COLUMN-I |                                                                                                                                                                | COLUMN-II |     |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----|
| (A)      | The number of 5 digit numbers having the product of digits 20 is                                                                                               | (p)       | 77  |
| (B)      | A man took 5 space plays out of an engine to clean then. The number of ways in which he can place at least two plays in the engine from where they came out is | (q)       | 30  |
| (C)      | The number of integers between 1 and 1000 inclusive in which at least two consecutive digits are equal is                                                      | (r)       | 50  |
| (D)      | The value of $\frac{1}{15} \sum_{1 \leq i \leq j \leq 9} i \cdot j$                                                                                            | (s)       | 181 |
|          |                                                                                                                                                                | (t)       | 31  |

A) A-R, B-T, C-S, D-P,

B) A-P, B-P,Q, C-QR, D-P,Q,R,S

C) A-P,T, B-Q,R, C-QR, D-P,Q,R,S

D) A-P,QR, B-P,Q, C-R,P, D-P,Q

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*Rough Work*

52. Match the following functions of Column– I with their corresponding fundamental periods from Column– II.

|     | Column – I                                                                  |     | Column – II     |
|-----|-----------------------------------------------------------------------------|-----|-----------------|
| (A) | $\frac{\cot \frac{x}{4} + \tan \frac{x}{4}}{1 + \tan \frac{x}{2} - \tan x}$ | (P) | $\pi$           |
| (B) | $\sin\left(\frac{\pi}{3} - x\right) \sin\left(\frac{\pi}{3} + x\right)$     | (Q) | $3\pi$          |
| (C) | $\frac{ \sin x  +  \cos x }{1 +  \sin x  +  \cos x }$                       | (R) | $\frac{\pi}{2}$ |
| (D) | $3\sin^5 x + \cos^3 x + 4\sin^7 x + \cos^4 x$                               | (S) | $4\pi$          |
|     |                                                                             | (T) | $2\pi$          |

A) A-P, B-Q, C-R, D-S

B) A-S, B-P, C-R, D-T

C) A-S, B-Q, C-R, D-P

D) A-S, B-R, C-P, D-Q

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*Rough Work*

53. Match the following:-

The value of  $\sqrt{1+\sin A} + \sqrt{1-\sin A}$  is given in I then the interval in which  $A/2$  lies in II

| COLUMN-I |                 | COLUMN-II |                                                 |
|----------|-----------------|-----------|-------------------------------------------------|
| (A)      | $2 \sin A / 2$  | (P)       | $\left[ \frac{3\pi}{4}, \frac{5\pi}{4} \right]$ |
| (B)      | $2 \cos A / 2$  | (Q)       | $\left[ \frac{5\pi}{4}, \frac{7\pi}{4} \right]$ |
| (C)      | $-2 \sin A / 2$ | (R)       | $\left[ \frac{-\pi}{4}, \frac{\pi}{4} \right]$  |
| (D)      | $-2 \cos A / 2$ | (S)       | $\left[ \frac{\pi}{4}, \frac{3\pi}{4} \right]$  |

A) A-P, B-Q, C-R, D-S

B) A-S, B-P, C-R, D-T

C) A-S, B-Q, C-R, D-P

D) A-S, B-R, C-Q, D-P

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*Rough Work*

54. Match the following:-

| COLUMN-I |                                                                                                                                                                                | COLUMN-II |   |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---|
| (A)      | If the binomial coefficients of the $r^{\text{th}}(r+1)^{\text{th}}$ and $(r+2)^{\text{th}}$ term in the expansion of $(1+x)^{14}$ are in A.P., then r can be equal to         | (P)       | 5 |
| (B)      | The sum of the coefficients in the polynomial expansion of $\left[ \left\{ 1+x+x^2+x^3+x^4+\dots+(x)^{n-1} \right\} (1-x) \right]^m, (n, m \in N)$ , is                        | (Q)       | 4 |
| (C)      | Sum of the series $\sum_{r=1}^n (-1)^r \cdot {}^nC_r \left[ 1+i^r+i^{2r}+\dots+i^{(n-1)r} \right]$ where $(n=4k, k \in I)$ is equal to ( where $i = \text{iota} = \sqrt{-1}$ ) | (R)       | 0 |
| (D)      | If $3^{37} = 80\lambda + k, \lambda \in N$ , then k is                                                                                                                         | (S)       | 3 |

A) A-P, B-Q, C-R, D-S

B) A-S, B-P, C-R, D-T

C) A-S, B-Q, C-R, D-P

D) A-P, B-R, C-R, D-S

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*Rough Work*